Course notes

Definition and properties

The exponential is continuous, strictly increasing on $\mathbb{R}$, and its image is $]0,+\infty[$. For every $x>0$, there is a unique real number y such that $e^y = x$: it is denoted $\ln x$.

$$\ln(e^y) = y \quad (y \in \mathbb{R}), \qquad e^{\ln x} = x \quad (x>0), \qquad \ln 1 = 0, \quad \ln e = 1.$$

For $a, b > 0$ and $n \in \mathbb{Z}$,

$$\ln(ab) = \ln a + \ln b, \quad \ln\left(\frac{a}{b}\right) = \ln a - \ln b, \quad \ln(a^n) = n \ln a, \quad \ln\sqrt{a} = \tfrac{1}{2}\ln a.$$

Proof for the product. $e^{\ln a + \ln b} = e^{\ln a} e^{\ln b} = ab$; the uniqueness of the preimage gives the formula. There is no similar formula for $\ln(a+b)$.

Derivative, variations and limits

We assume that ln is differentiable on $]0,+\infty[$. Differentiating $e^{\ln x} = x$ gives $e^{\ln x}(\ln x)' = 1$; so $(\ln x)' = \frac{1}{x} > 0$. Ln is strictly increasing and concave, because its second derivative is $-\frac{1}{x^2}$.

$$\lim_{x\to 0^+} \ln x = -\infty, \qquad \lim_{x\to+\infty} \ln x = +\infty.$$

Its curve is the reflection of that of the exponential in $y = x$. The line $x = 0$ is a vertical asymptote. Its tangent at 1 gives ln $x \leq x - 1$ for $x > 0$.

Composition and comparative growth

If $u > 0$ is differentiable, $(\ln u)' = \frac{u'}{u}$. For every integer $m \geq 1$,

$$\lim_{x\to+\infty} \frac{\ln x}{x^m} = 0, \qquad \lim_{x\to 0^+} x^m \ln x = 0.$$

Proof at 0. Let $t = -\ln x$; when $x \to 0^+$, $t \to +\infty$ & $x \ln x = -te^{-t} \to 0$ by comparative growth. For $m \geq 1$, the same reasoning gives $-te^{-mt} \to 0$. At $+\infty$, set $t = \ln x$ to write $\frac{\ln x}{x^m} = \frac{t}{e^{mt}} \to 0$.

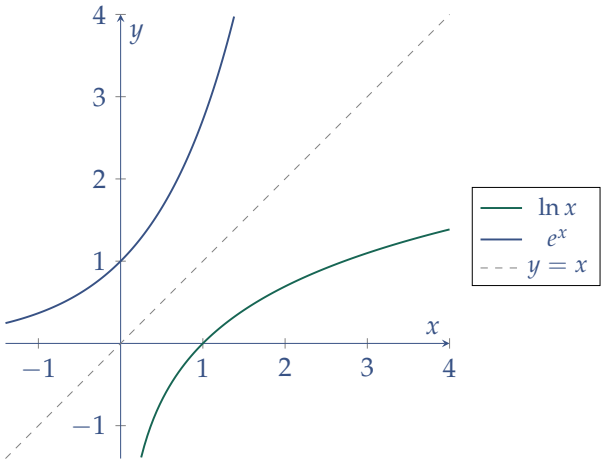
Equations and thresholds

For $A, B > 0$, $\ln A = \ln B \Leftrightarrow A = B$ and $\ln A \leq \ln B \Leftrightarrow A \leq B$. Always start with the domain.

Example. $\ln(x-1) + \ln(x+1) = \ln 8$ requires $x > 1$. The equation becomes $x^2 - 1 = 8$, so $x = 3$ only.

Bound. $0{,}8^n \leq 0.1$ is equivalent to $n \ln 0{,}8 \leq \ln 0.1$, then $n \geq \frac{\ln 0{,}1}{\ln 0{,}8} \simeq 10.319$. The first integer is 11.

Worked study. For $f(x) = \frac{\ln x}{x}$, $f' = \frac{1-\ln x}{x^2}$; f increases on $]0,e]$, decreases on $[e,+\infty[$, and its maximum is $\frac{1}{e}$. Its limit at 0 $^+$ is $-\infty$, its limit at $+\infty$ is 0.



Use the logarithm with its conditions

6.1 The inverse of the exponential

First use: S44.

Definition · Natural logarithm

For $x > 0$, $\ln x$ is the unique real number y such that $e^y = x$. Thus $e^{\ln x} = x$ for $x > 0$ and $\ln(e^y) = y$ for any real y. The logarithm transforms a positive value into a real exponent.

Worked example · Domain before transforming

To solve $\ln(x-1) = \ln(3-x)$, both arguments must be positive, so $1 < x < 3$. The fact that ln is strictly increasing then gives $x - 1 = 3 - x$, therefore $x = 2$, which belongs to the domain.

A1 · Immediate application Determine the domains of $\ln(2-x)$ and of $\ln(x^2-1)$. Solve $\ln(2x+1) = 0$, stating the domain.

6.2 Products, quotients and powers

First use: S46.

Property · Algebraic rules

For $a, b > 0$, $\ln(ab) = \ln a + \ln b$ and $\ln(a/b) = \ln a - \ln b$. For an integer m, $\ln(a^m) = m \ln a$. The conditions on the arguments must still be checked after a transformation.

Proof. Let $u = \ln a$, $v = \ln b$. Then $ab = e^u e^v = e^{u+v}$, hence $\ln(ab) = u + v$. The quotient comes from $\ln(b \times b^{-1}) = \ln 1 = 0$, and that of powers by repeating the product, then taking reciprocals. □

Worked example · An exact expression

$\ln 12 - \ln 3 = \ln 4 = 2 \ln 2$. On the other hand $\ln(1+1) = \ln 2 \neq \ln 1 + \ln 1 = 0$. The logarithm is not distributive over addition.

A2 · Two mistakes to explain Disprove " $\ln(a+b) = \ln a + \ln b$ ". Then explain why $\ln(x^2) = 2 \ln x$ is not an identity on $\mathbb{R} \setminus \{0\}$ and give the correct identity on this set.

6.3 Studying a logarithmic function

First use: S47.

Worked example · Variation and maximum

On $]0; +\infty[$, $f(x) = \frac{\ln x}{x}$ has derivative $f'(x) = \frac{1 - \ln x}{x^2}$. It increases up to $x = e$, then decreases, and its maximum is $1/e$. Its sign depends on $\ln x$, so it is negative before 1; this does not prevent it from increasing on $]0; 1[$.

A3 · Partial guidance Study the domain, derivative and sign of $g(x) = \ln(1+x^2)$. Explain why the function is even and what this implies for its variations.

6.4 The sign of $\ln q$ determines the inequality

First use: S50.

Worked example · A threshold for a decreasing sequence

For $n \in \mathbb{N}$, $0{,}8^n < 0{,}1$ is equivalent to $n \ln 0{,}8 < \ln 0{,}1$. Since $\ln 0{,}8 < 0$, dividing reverses the order: $n > \ln 0{,}1 / \ln 0{,}8 \simeq 10{,}32$. The first index is 11; $0{,}8^{10} \simeq 0{,}1074$ and $0{,}8^{11} \simeq 0{,}0859$ check the boundary. For $q > 1$, $\ln q > 0$ and the order is preserved; for $q = 1$, we do not divide by $\ln q = 0$.

A4 · Choose the right direction Determine the smallest integer $n \geq 0$ such that $1{,}05^n \geq 2$, then such that $0{,}9^n \leq 0.2$. Check the previous index in each case.

6.5 Comparing and changing variable

First use: S49.

Property · Standard limits

$\ln x \to +\infty$ at $+\infty$ & $\ln x \to -\infty$ at 0^+. For every integer $m \geq 1$, $(\ln x)/x^m \to 0$ at $+\infty$; also $x \ln x \to 0$ at 0^+. The first comparative growth result follows from $t/e^{mt} \to 0$ with $t = \ln x$.

Worked example · Adapting a comparative growth result

At $+\infty$, $\frac{2+\ln x}{x} = 2/x + \ln x/x \to 0$. At 0^+, $x^2 \ln x$ becomes $-te^{-2t}$ with t=-ln x; this expression tends to 0.

A5 · Deferred application Calculate the limits of $\ln(2x)/x$ at $+\infty$ and of $x \ln(3x)$ at 0^+. Show a transformation using algebraic rules.

A6 · Synthesis problem For each real a, determine the number of positive solutions of $\ln x = ax$. Use the study of x/x, its limits and the position of the level a.

A first step: Distinguish $a < 0$, $a = 0$, $0 < a < 1/e$, $a = 1/e$ and $a > 1/e$. A limit that is not attained does not give an additional solution.

One minute, spoken aloud. Explain why "taking logarithms" is not allowed on a expression whose positivity has not been verified.

Delayed review · S58 then S66. Identify a cooling constant, then find a antiderivative of $1/x$ keeping the interval of definition.

Investigation

Checking a calculation error - 30 min.

A calculation method produces an initial error $e_0 = 5$. At each step, the error is multiplied by 0.7. Let e_n be error after n steps, for $n \in \mathbb{N}$.

1. Express e_n as a function of n.

2. Using logarithms, determine the smallest integer n such that $e_n \leq 10^{-4}$.

3. Calculate, with a calculator, the error at the index found and at the previous index to check minimality of the index.

4. We now require an error at most equal to 10^{-6}. Determine the new minimum number of steps. Does dividing the tolerance by 100 multiply the number of steps by 100?